LOGISTIC REGRESSION

```
import numpy as np
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt

X = np.array([[1], [2], [3], [4], [5], [6], [7], [8]])
y = np.array([0, 0, 0, 0, 1, 1, 1, 1])

model = LogisticRegression()
model.fit(X, y)

hours = np.array([[6]])
print("Prediction:", model.predict(hours))

plt.scatter(X, y)
plt.plot(X, model.predict_proba(X)[:, 1], color='red')
plt.xlabel("Hours Studied")
plt.ylabel("Probability")
plt.show()
```